Artificial Intelligence Review (2024) 57:330
https://doi.org/10.1007/s10462-024-10972-3

The survey on the dual nature of xAI challenges in intrusion detection and their potential for AI innovation

Marek Pawlicki[1,2] · Aleksandra Pawlicka[1,3] · Rafał Kozik[1,2] · Michał Choraś[1,2]

Accepted: 24 September 2024 / Published online: 16 October 2024
© The Author(s) 2024

Abstract
In the rapidly evolving domain of cybersecurity, the imperative for intrusion detection systems is undeniable; yet, it is increasingly clear that to meet the ever-growing challenges posed by sophisticated threats, intrusion detection itself stands in need of the transformative capabilities offered by the explainable artificial intelligence (xAI). As this concept is still developing, it poses an array of challenges that need addressing. This paper discusses 25 of such challenges of varying research interest, encountered in the domain of xAI, identified in the course of a targeted study. While these challenges may appear as obstacles, they concurrently present as significant research opportunities. These analysed challenges encompass a wide spectrum of concerns spanning the intersection of xAI and cybersecurity. The paper underscores the critical role of xAI in addressing opacity issues within machine learning algorithms and sets the stage for further research and innovation in the quest for transparent and interpretable artificial intelligence that humans are able to trust. In addition to this, by reframing these challenges as opportunities, this study seeks to inspire and guide researchers towards realizing the full potential of xAI in cybersecurity.

Keywords Challenges · Cybersecurity · Explainability · Intrusion detection · xAI

1 Introduction

In our current era, characterized by the profound interconnectivity enabled by the Internet and its attendant services, the consistent threat from adversaries is undeniable. Ensuring the integrity and security of the networks mandates the employment of intrusion detection systems (IDS). The issues faced by these systems are intensified by the burgeoning complexity of networks and the sophistication of adversarial strategies. As it has been established, the

✉ Michał Choraś
chorasm@pbs.edu.pl

1 ITTI, Poznań, Poland

2 Bydgoszcz University of Science and Technology, Bydgoszcz, Poland

3 University of Warsaw, Warsaw, Poland

efficacy of these systems in real-time threat recognition could be significantly augmented by the integration of machine learning (ML) and deep learning (DL) methodologies (Mijwil et al. 2023; Pawlicki et al. 2022).

However, while contemporary algorithms demonstrate incredible accuracy and expedited response times, they frequently exhibit a notable opacity in their decision-making processes, rendering them not understandable to human operators (Yan et al. 2022). This lack of transparency, often referred to as the "black box" nature of AI (Haque et al. 2023), is especially concerning in high-stakes sectors such as cybersecurity, healthcare, and law enforcement, where decisions may have profound and irreversible impacts on the lives and welfare of individuals (Capuano et al. 2022; Nwakanma et al. 2023). The inability to interpret how these algorithms arrive at their conclusions creates significant challenges, including difficulty in troubleshooting errors, identifying biases, and ensuring fairness and accountability in automated decisions.

To address these challenges, the paradigm of Explainable Artificial Intelligence (xAI) has emerged as a critical area of research and application (Choraś et al. 2020; Ali et al. 2023). xAI aims to create models and methods that make the decision-making processes of AI systems more transparent and interpretable to human users. At its core, explainability seeks to deconstruct the internal mechanics of an algorithm, providing insights into how inputs are processed to yield outputs, which is essential for building trust and understanding between AI systems and human operators.

Explainability in AI is achieved through a variety of methods, broadly categorized into post-hoc explanations and inherently interpretable models, also known as ante-hoc explanations (Retzlaff et al. 2024). Post-hoc explanations refer to techniques applied after a model has made a decision, aiming to provide insights into how that decision was reached. Common post-hoc techniques include LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations), which help to interpret complex models by approximating their behaviour locally. LIME works by perturbing the input data and observing changes in the output to build a simpler, interpretable model that mimics the original model's behaviour in the vicinity of the input. This allows analysts to understand how specific features influence the model's predictions for individual cases (Ribeiro et al. 2016). SHAP, on the other hand, is based on cooperative game theory and provides a unified measure of feature importance by calculating the contribution of each feature to the prediction across all possible feature combinations (Lundberg and Lee 2017). These techniques are particularly useful in domains like cybersecurity, where understanding the contribution of various features, such as network traffic patterns or user behaviour, is critical for effective threat detection and response (Sarker et al. 2024).

In contrast, inherently interpretable models are designed to be transparent from the outset. Examples include decision trees, linear models, and rule-based systems. These models are structured in a way that their decision-making process is straightforward and easily understood by humans. For instance, decision trees operate by splitting the data based on feature values, leading to a tree-like structure where the path from root to leaf represents the decision logic (Saha and Manickavasagan 2021). This clear visual representation of how decisions are made makes decision trees particularly useful in situations where transparency is a priority, such as in cybersecurity, where decision trees can be employed to create transparent rules for identifying and classifying threats.

In the domain of cybersecurity, the addition of explainability through these techniques is particularly valuable. It allows human analysts to better understand the rationale behind AI-driven threat detection and mitigation strategies, enabling them to formulate more informed and effective countermeasures (Sharma et al. 2022). For example, by applying explainability to a complex model that detects anomalies in network traffic, analysts can identify which specific network features led the model to flag certain activities as suspicious. This insight helps validate the model's predictions and enhances the analysts' ability to fine-tune the model and the overall security protocols. In addition to this, xAI can enhance the overall efficacy and fairness of AI systems by highlighting biases and inaccuracies, which is crucial in environments where errors can have severe consequences. Identifying and correcting biases is essential to ensure that the AI system operates fairly and does not inadvertently target or overlook specific groups, thereby improving the reliability and trustworthiness of AI-driven decisions in critical security operations (Zhang et al. 2022). Lastly, applying explainability helps ensure that even when complex models are necessary for high accuracy, their decisions remain transparent and understandable.

The importance of xAI extends beyond merely interpreting algorithmic decisions; it also plays a crucial role in ensuring compliance with legal and ethical standards. For instance, regulations like the General Data Protection Regulation (GDPR) in the European Union (Pawlicka et al. 2020) emphasise the right to explanation, requiring that individuals affected by automated decisions be provided with meaningful information about the logic involved (Goodman and Flaxman 2017). In such regulatory environments, xAI becomes not just a technical necessity but a legal obligation, further emphasising its significance in contemporary AI applications.

The concept of xAI has been illustrated in Fig. 1 which demonstrates the process through which AI models can be made more transparent and interpretable.

Researchers, such as Meske and colleagues, have highlighted that the concept of explainability is not merely an emergent concern aligned with the progression of AI. Rather, it is a foundational issue that has been present since AI's very inception (Meske et al. 2022). However, it has also been underscored that despite this historical perspective, the world of explainability is still in its formative stages, leaving substantial room for future exploration and inquiry (Mathew 2023). This means that the field has yet to overcome a number of challenges and issues. On account of this, the authors of this paper wished to study which exact challenges xAI is facing that could potentially alter the course of its development. Special emphasis had been put on the issues relevant to cybersecurity. Yet, these challenges, as this paper emphasises, can at the same time become gateways to deeper understanding and innovative solutions.

In the course of the study, the authors were able to pinpoint 25 challenges of varying research interest. The challenges have been organised from the most often mentioned to the

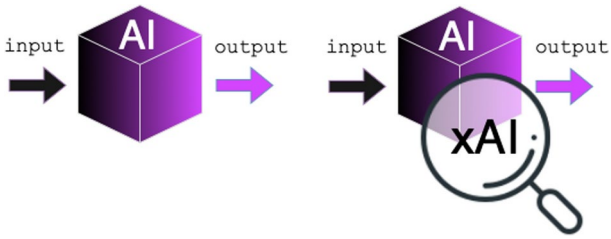

Fig. 1 On the left: a "black-box" AI model; on the right: the inner workings of a model made transparent with xAI

more obscure ones. Each of them has been accompanied by the discussion of the potential opportunities they might create if approached as such.

1.1 Contribution

As it turned out, while the challenges to xAI might seem intimidating, they concurrently offer avenues for research and exploration. As this paper has adopted an open approach, it deems no issue to xAI in cybersecurity solely as an obstacle; rather, all of the presented issues come with their possible advantages outlined. Thus, the resulting comprehensive collection may serve both as a set of guidelines on what to be careful of when considering xAI, and as an inspirational repository for those aiming to redefine these challenges into innovative solutions.

1.2 Paper organisation

The remainder of the paper is organised as follows: Sect. 2 discusses the materials and methods applied in the study. The following Sect. 3 presents the outcomes of the study, whilst Sect. 4 concludes the paper. This structure of the work has been reflected in the roadmap presented in Fig. 2.

2 Materials and methods

In this section, the details concerning the study presented have been discussed, such as the process of posing research questions, constructing the search string, and the overall course of the study.

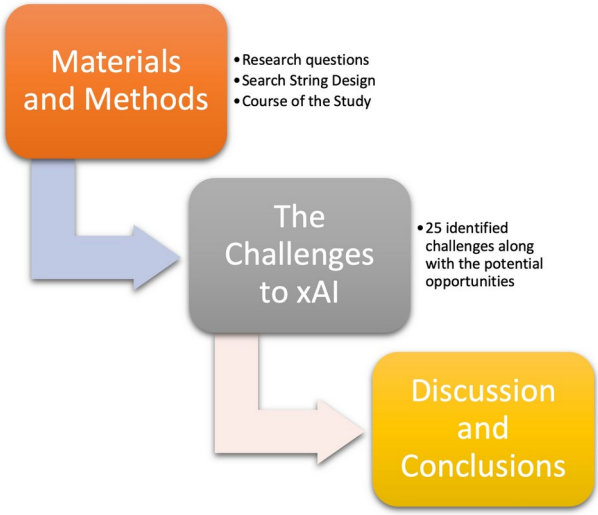

Fig. 2 The roadmap for this paper

2.1 Research questions

Given the rationale presented in the Introduction, the authors of this paper wished to find out what the challenges for the application of xAI are. Preferably, they wanted to uncover the ones of relevance to cybersecurity/network intrusion detection. The adopted open approach let the authors anticipate that the uncovered issues can be at the same time presented as potential opportunities, or could be seen as one without being directly named as such.

In order to address these scientific concerns, two research questions have been crafted using the PICO methodology (Yensen 2013; Eldawlatly et al. 2018). The PICO framework is a structured approach traditionally used in evidence-based research, which has been adapted for various fields; it aids in clearly formulating research questions and guiding systematic reviews by breaking down the essential components of the study. PICO stands for Problem (also dubbed Population or Patient, depending on the specific field it is applied to), Intervention, Comparison, and Outcome. The "P" refers to the specific issue or challenge being addressed, the "I" describes the strategy implemented to address the identified problem, and "C" involves identifying a control or baseline condition to evaluate the effectiveness of the Intervention. Lastly, "O" specifies the expected results or impacts of the Intervention. Yensen (2013) remarks that in most cases it is enough just to specify the P" and "I" (Yensen 2013). Thus, the authors have decided that with the first issue, the Population (or "Problem") is the application of explainable AI within the field of cybersecurity, specifically in network intrusion detection, the Intervention is the implementation or use of xAI technologies and the Outcome equals understanding the challenges faced in the application of xAI in this specific field. Consequently, for the other issue, with the Population, the focus was on xAI technologies in cybersecurity, specifically how challenges in this area might drive innovation, the Intervention could be considered as addressing or leveraging the identified challenges and the Outcome here would be the identification of potential opportunities for innovation in AI. The resulting Research Questions (RQ) were:

RQ1: What are the challenges for the application of xAI (in cybersecurity/network intrusion detection)?

RQ2: What potential opportunities for AI innovation arise from these challenges?

The answers to both questions have been discussed in Sect. 3.2.2 Search string design

In order to answer the Research Questions posed, for the sake of this study, the method of a targeted literature review has been employed. It consisted in the systematic exploration of existing academic literature to gather and synthesize relevant knowledge (Snyder 2019).

After the research question has been posed, the search string was developed; in this case the principles of the PICO methodology (Yensen 2013) have been applied as well. The resulting string was as follows:

(('xai' OR 'explainable AI' OR 'explainability') AND ('intrusion detection' or 'cybersecurity') AND 'challenge' OR 'issue' OR 'problem'). Subsequently, the string was used for searching the following databases:

- IEEE Xplore
- DBLP

- Science Direct
- arXiv
- ResearchGate.Such a choice of sources was meant to ensure the review of both established and emerging research across a wide spectrum of scientific publications was adequately comprehensive.

2.3 Course of the study

The initial search yielded 1532 results. Subsequently, the inclusion and exclusion criteria were applied. The criteria have been presented in Table 1.

The rationale for employing specific inclusion and exclusion criteria in the research study was crucial for ensuring the relevance, quality, and recency of the information gathered. The criterion concerning including only peer-reviewed papers based on the conviction that peer review is a fundamental process in scholarly publishing that ensures the validity, significance, and originality of academic work, and including only peer-reviewed papers helps ensure that the data and findings are examined and validated by experts in the field before publication, enhancing the credibility of the study. Limiting the search to English-language papers was practical for the researchers, it also made the study more accessible to a wider international audience. Focusing on recently published papers ensured that the study incorporates the latest findings and state-of-the-art knowledge in the rapidly evolving areas of cybersecurity and AI. While there are scenarios where historical data can still be highly relevant or valuable (For instance, historical data can serve as a baseline for detecting emerging trends or anomalies when juxtaposed with current data, etc.), older data might sometimes not reflect current trends or technologies, particularly in rapidly evolving fields like cybersecurity. Including only available resources ensured that the research could be verified and replicated by other scholars. Lastly, excluding duplicates was essential to prevent skewing the results by counting the same data multiple times.

It must be noted that the authors agreed to override the exclusion criteria if they found a work of exceptional quality and/or relevance to the study presented. The possibility of overriding the exclusion criteria introduces flexibility into the systematic review process. This is significant because it allows for the inclusion of standout works that might not meet one or more of the specified criteria but are deemed highly relevant or innovative. Such flexibility can be crucial in fields like AI and cybersecurity, where groundbreaking research could be published in less conventional formats or venues. It also underscores the importance of expert judgement in the review process, allowing researchers to use their discretion to include studies that might significantly impact the field or the particular research question, even if they do not fit the standard criteria.

After the application of the aforementioned criteria, and screening the titles and abstracts, the number was narrowed down to 76. The remaining papers were then subjected to a

Table 1 The inclusion and exclusion criteria for the study

Criterion	Inclusion	Exclusion
Peer-reviewed scientific paper	Yes	No
Labguage: English	Yes	No
Published in or after 2018	Yes	No
Availabilty	Yes	No
Duplicate	No	Yes

more in-depth analysis, where each paper was evaluated based on several additional factors, including the clarity of the research methods used, the relevance and specificity of the findings to the research questions, and the recency of the research to ensure the inclusion of up-to-date information. Furthermore, papers that provided unique insights or addressed gaps not covered by other studies were prioritised. Through this thorough review process, 28 papers were ultimately selected as the most relevant and appropriate for inclusion in the study. These papers were then carefully analysed in detail, and the results of this analysis have been presented in Table 2 and discussed in the following sections.

Through a comprehensive analysis of the 28 papers selected for the study, 25 unique challenges (presented in Table 2) had been identified. The challenges were ranked based on their frequency of occurrence across these papers, with those mentioned most frequently appearing at the top of the list, indicating their prominence and perceived importance within the field. Challenges that were mentioned less frequently but still found to be relevant were included towards the end of the list. Such an approach was adopted to ensure that the list reflects both the most widely recognized challenges as well as those that, while less commonly discussed, are still significant within specific contexts.

3 The challenges to xAI

In this section, 25 potential challenges to the adoption and development of explainability have been discussed that had been identified in the course of the presented study, as shown and organised in Table 2. The challenges have been arranged from the ones the subject literature mentions the most to the least-discussed, more obscure ones. Fig. 3 illustrates the proportional representation of these challenges in the literature. In addition, the framing of these challenges or issues has been intentionally reversed; rather than viewing them solely as obstacles, they are presented as potential opportunities for innovation and improvement in the field. This approach highlights how each challenge can be seen as a catalyst for developing new strategies and enhancements in explainability.

3.1 Standardisation challenges in defining explainability

The first challenge to confront stems from the absence of a uniform characterization of explainability within the domain of xAI. This underscores the imperative for a universal blueprint that both researchers and technical experts might utilize in the formulation and architecture of xAI systems. Evaluating current studies in a methodical manner presents inherent complexities. Crafting this overarching schema necessitates a multidimensional examination of explainability, encompassing aspects like data sources, model structures, learning methodologies, resultant outputs, and gradient data insights (Ding et al. 2022).

The analysed works of a number of scientists concern the imperative task of establishing a comprehensive understanding of explainability in xAI. The xAI research community has made substantial strides in exploring explainability from diverse perspectives, yet consensus on its definition remains hardly possible to reach (Ding et al. 2022); the absence of a common foundational definition hampers the development and design of xAI solutions.

To advance xAI, there is a growing call for increased formalism and quantification of explanations (Saeed and Omlin 2023). Existing works often focus on problem formula-

tions that may become irrelevant as interpretation methods vary. A rigorous and generalized approach to the expansibility problem is essential to improve the identification, classification, and evaluation of explainability sub-issues (Saeed and Omlin 2023). Additionally, formalizing the theoretical aspects of xAI is crucial to answer fundamental questions, such as the weighting of model and data distribution in generated explanations. Beyond formal definitions, the xAI research must also address the credibility of explanations, especially in critical situations (Ali et al. 2023). In order to confirm the reliability of xAI tools when deployed in practical applications, methodological measures and assurances of transparency are essential. Intrusion detection systems represent a critical domain where xAI can have a significant impact. However, defining explainability in the context of IDS is a challenge due to the lack of consensus, too (Neupane et al. 2022). Using foundational xAI definitions while incorporating security domain-specific elements, such as the principles of confidentiality, integrity and availability (Pawlicki et al. 2023), contributes to a more concrete understanding.

It must also be mentioned that the concept of interpretability itself is subjective (Antoniadi et al. 2021). Diverse techniques, including user studies and experiments, have been employed to address this subjectivity. However, there remains a need for a generic method to evaluate explanations consistently. The pursuit of meaningful progress in xAI, particularly in the cybersecurity domain, requires a unified view of explainability (Javed et al. 2023). Fragmented approaches in application areas, such as Malware Detection, should be consolidated to provide a clearer and more coherent perspective.

Potential opportunities The lack of a standard for explainability in xAI invites the opportunity for AI to innovate through the creation of a universal framework, promoting interdisciplinary collaboration and establishing robust, field-specific standards. Addressing the subjectivity of interpretability opens the door for AI to develop adaptable, user-centric explanations, enhancing reliability and trust across critical applications, including cybersecurity.

3.2 Balancing complexity and interpretability a.k.a. performance vs. explainability trade-off

A number of researchers notice that xAI is faced with another crucial dilemma: choosing between the simplicity of white-box models, which are easier to understand but less accurate, and the complexity of black-box models, which offer high accuracy but are difficult to interpret (Ding et al. 2022). Thus, they worry about the potential trade-off between model complexity and interpretability within the context of ML and AI.

Complex models are often presumed to yield superior accuracy, but this assumption is not universally valid (Ding et al. 2022). There occur situations where model complexity may or may not translate into improved accuracy. Furthermore, studies investigate how explainability methods impact this trade-off, highlighting that the performance of xAI algorithms can degrade as explanation techniques become more intricate.

As Hulsen puts it, the choice between a 'glass box' model, which prioritises explainability, and a 'crystal ball' model, which emphasises accuracy, is a central theme in xAI (Hulsen 2023). Researchers have also examined instances where achieving full explainability may necessitate simplifying the model, potentially sacrificing accuracy. Alternatively, introducing additional transparency-enhancing steps may also impact performance.

Table 2 The identified challenges to xAI, as mentioned in the subject literature

		Yang et al. (2023)	Zhou and Hu (2023)	Brik et al. (2023)	Yurrita et al. (2023)	Amirian et al. (2023)	Swamy et al. (2023)	Hulsen (2023)	Javed et al. (2023)	Saeed and Omlin (2023)	Ali et al. (2023)	Machlev et al. (2022)	Srivastava et al. (2022)	Neupane et al. (2022)	Nadeem et al. (2022)
1	Standardisation Challenges in Defining Explainability			X					X		X	X		X	X
2	Performance vs. Explainability Trade-off	X		X		X		X				X		X	X
3	Tailoring Explanations to Users Expertise	X		X			X		X	X		X		X	
4	Privacy and Security			X				X	X	X	X		X		X
5	Developing Evaluation Metrics	X						X	X	X		X		X	X
6	Increasing Complexity, Systematic Instability and Multiplicity of Explanations	X	X	X		X		X							
7	Adversarial Use of xAI							X		X		X	X		X
8	User-Centric Validation and Evaluation	X		X				X	X	X					X
9	Usability and Interpretability	X		X				X	X						

Table 2 (continued)

		Yang et al. (2023)	Zhou and Hu (2023)	Brik et al. (2023)	Yurrita et al. (2023)	Amirian et al. (2023)	Swamy et al. (2023)	Hulsen (2023)	Javed et al. (2023)	Saeed and Omlin (2023)	Ali et al. (2023)	Machlev et al. (2022)	Srivastava et al. (2022)	Neupane et al. (2022)	Nadeem et al. (2022)
10	Interdisciplinary Collaboration			X	X					X	X	X			
11	Time-Sensitive Explanations	X		X			X						X		
12	Ensuring Fairness, Accountability, and Trustworthiness									X	X		X		
13	Quality Natural Language Explanations									X					
14	Data Quality	X				X				X					
15	Explainability for Diverse Data Types									X			X		
16	Information Gap and Confirmation Bias									X					
17	Evaluating and Optimizing xAI Visualizations and Interpretability Methods								X						
18	Ensuring Fidelity and Causality														

Table 2 (continued)

		Yang et al. (2023)	Zhou and Hu (2023)	Brik et al. (2023)	Yurrita et al. (2023)	Amirian et al. (2023)	Swamy et al. (2023)	Hulsen (2023)	Javed et al. (2023)	Saeed and Omlin (2023)	Ali et al. (2023)	Machlev et al. (2022)	Srivastava et al. (2022)	Neupane et al. (2022)	Nadeem et al. (2022)
19	Maintaining Accurate Representations of AI Capabilities														
20	Balancing the Benefits and Risks of Incorporating Explainability into AI Systems														
21	xAI Methods for Tabular Data														
22	Challenges in Model and Method Development									X					
23	Contrastive and Counterfactual Explanations									X					
24	Uncertainty Quantification									X					
25	Economic Considerations									X					

Table 2 (continued)

		Capuano et al. (2022)	Jagatheesaperumal et al. (2022)	Kalyanathaya (2022)	Islam et al. (2022)	Evans et al. (2022)	Ding et al. (2022)	Korica et al. (2021)	Antoniadi et al. (2021)	Sovrano et al. (2021)	Paredes et al. (2021)	Kabir et al. (2021)	Tjoa and Guan (2021)	Adadi and Berrada (2018)	Lipton (2016)
1	Standardisation Challenges in Defining Explainability	X		X	X		X	X	X						
2	Performance vs. Explainability Trade-off						X			X			X	X	X
3	Tailoring Explanations to Users Expertise	X		X	X				X						
4	Privacy and Security					X	X					X			
5	Developing Evaluation Metrics				X		X	X							
6	Increasing Complexity, Systematic Instability and Multiplicity of Explanations			X			X							X	
7	Adversarial Use of xAI	X					X	X							
8	User-Centric Validation and Evaluation	X							X						
9	Usability and Interpretability	X					X	X				X			
10	Interdisciplinary Collaboration								X					X	
11	Time-Sensitive Explanations														
12	Ensuring Fairness, Accountability, and Trustworthiness														
13	Quality Natural Language Explanations				X						X				
14	Data Quality														
15	Explainability for Diverse Data Types		X												
16	Information Gap and Confirmation Bias					X									
17	Evaluating and Optimizing xAI Visualizations and Interpretability Methods						X								
18	Ensuring Fidelity and Causality						X					X			

Table 2 (continued)

		Capuano et al. (2022)	Jagatheesaperumal et al. (2022)	Kalyanathaya (2022)	Islam et al. (2022)	Evans et al. (2022)	Ding et al. (2022)	Korica et al. (2021)	Antoniadi et al. (2021)	Sovrano et al. (2021)	Paredes et al. (2021)	Kabir et al. (2021)	Tjoa and Guan (2021)	Adadi and Berrada (2018)	Lipton (2016)
19	Maintaining Accurate Representations of AI Capabilities					X							X		
20	Balancing the Benefits and Risks of Incorporating Explainability into AI Systems					X									
21	xAI Methods for Tabular Data						X								
22	Challenges in Model and Method Development														
23	Contrastive and Counterfactual Explanations														
24	Uncertainty Quantification														
25	Economic Considerations														

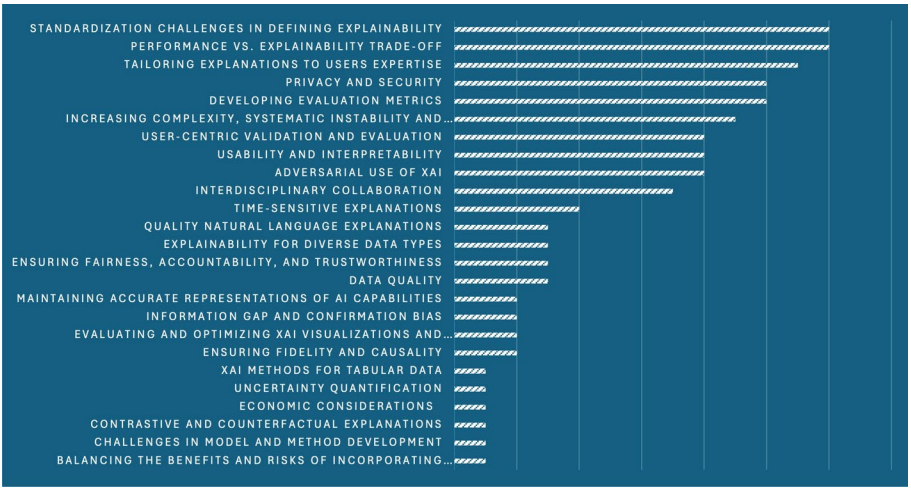

Fig. 3 Proportional representation of the 25 challenges based on their frequency in the analysed literature

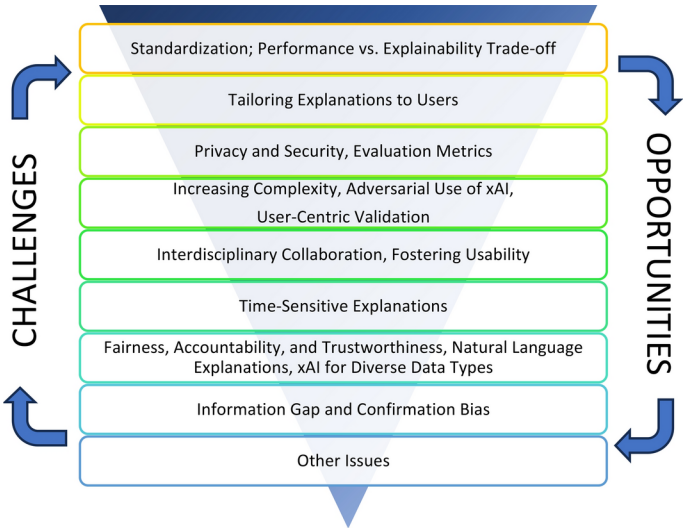

Fig. 4 The challenges to/the opportunities of the application of xAI in cybersecurity as identified in the course of the study presented; the challenges have been organised to reflect the degree of scientific interest expressed by how often the issue has been acknowledged in the studied literature

Another crucial consideration in xAI, especially in cybersecurity, is the 'price of interpretability' (Nadeem et al. 2022). In the specific context of IDS, performance is paramount, as delays can result in losses for Cyber Security Operation Centres (Neupane et al. 2022). The optimization of explainable IDS with a focus on explainability have to address the need to balance explanation generation for every sample with strategic selection for explanations to prevent unnecessary performance slowdowns. The challenge of ensuring that interpretable surrogate models are certifiably equivalent to their black-box counterparts for

meaningful evaluations has to be explored. While post-hoc explainability is prevalent, the researchers suggest examining the limited adoption of interpretable models and its implications. The relationship between robustness and interpretability is another intriguing aspect (Nadeem et al. 2022). Initial research suggests that robust models may inherently offer greater interpretability, warranting further investigation into this nexus.

Potential opportunities The dilemma between model simplicity and performance in xAI offers an opportunity for AI to progress by exploring hybrid models that balance understandability with high accuracy, potentially reshaping the assumed trade-off between complexity and interpretability. In cybersecurity, where speed is crucial, this challenge encourages the development of efficient, explainable systems that selectively generate explanations without compromising performance, thus advancing the field in creating robust yet interpretable AI solutions.

3.3 Tailoring explanations to user expertise and understanding

In the field of xAI, user diversity plays an important role as different users, ranging from data scientists and domain experts to decision-makers and non-experts, interact with machine learning models. Thus, researchers explore the significance of adapting explanations to cater to the backgrounds, knowledge levels, and communication styles of these users.

The user's background and proficiency play a significant role in shaping the cognitive structures they hold, and the depth of explanation they anticipate. At present, a multitude of explanation methodologies are designed predominantly for users with the capacity to decode explanations, leveraging their familiarity with the machine learning paradigm.

However, the importance of providing user-friendly explanations, enabling users with less technical knowledge to comprehend, is emphasised. Therefore, identifying the end-users and their goals becomes an important step in determining which aspects of the model require explanation (Kalyanathaya 2022).

To harness the knowledge of prior users effectively in predictive models, it is essential to come up with processes for digitally capturing and quantifying what and how much they knew before. This approach ensures that historical user insights can contribute to refining future explanations and model performance. Different users in diverse contexts should receive explanations that suit their specific needs and abilities. Clearly defining the objectives of users, as well as the goals of systems and explanations is essential, necessitating stakeholder involvement and thoughtful design of the system. On top of that, desirable explanations must not only be clear but also resistant to misinterpretation. This refers to situations where users can misinterpret explanations, leading to situations where the model is correct, but the user's interpretation is erroneous. Modifying the explainer to prevent user errors becomes essential in such cases (Neupane et al. 2022).

In the context of CSoCs, misclassified or incorrect explanations can have severe consequences. Analysts ought to analyse the reasoning underlying predictions in a critical way. On top of that, it is necessary to develop techniques for auditing the explanations which proved incorrect in the past (Neupane et al. 2022). In an ideal situation, an explainable IDS ought to be capable of auditing itself, as well as providing explanations for the audit.

Evaluating explanations poses a substantial challenge due to the varying levels of user expertise and understanding. More scientific effort is required to develop a methodology for evaluating explanations that accounts for users' intended expertise and capacity (Islam et

al. 2022). Different stakeholders may focus on distinct aspects of explanations, with some requiring a high-level overview while others demand in-depth details (Javed et al. 2023). Balancing these diverse needs is crucial.

Finally, as stated by Capuano, at the core of effective xAI, particularly in cybersecurity applications, there is always the human–machine interaction (Capuano et al. 2022). Building models that are understandable to humans is imperative, and it may involve supplementary responses to user queries to establish a comprehensive understanding. For example, Kalyanathaya et al. raise the question of how to convey machine learning model decisions in real-world terms to various stakeholders, emphasising the need for effective communication and bridging the gap between technical and non-technical users (Kalyanathaya 2022).

Potential opportunities The challenge of varying user expertise in xAI highlights the opportunity to develop adaptive explanation systems, which tailor information complexity to user understanding, thereby improving AI accessibility and usability across diverse user groups. This necessity for user-specific explanations in critical fields like cybersecurity also emphasises the importance of creating self-auditing, robust AI systems that not only explain their reasoning but also learn from past inaccuracies to refine their interpretability and ensure alignment with human cognition.

3.4 Privacy and security considerations

In the expansive digital world, data from diverse sources is omnipresent, giving rise to pronounced concerns regarding privacy during both its storage and transit phases. When confronted with xAI algorithms, these concerns intensify, compelling a need to strike an equilibrium between algorithmic transparency and the preservation of privacy. The intrinsic nature of xAI, aiming for clarity and interpretability, highlights an inherent tension between explainability and privacy considerations, as well as the potential privacy-explainability trade-offs.

AI algorithms often grapple with private data, and it is imperative to uphold individuals' privacy rights (Ding et al. 2022). Algorithms within the xAI domain, constructed with interpretability (or 'white-box' design) in mind, raise profound concerns regarding the potential threats to data privacy during both training and inference. This warrants further investigation, ensuring that xAI algorithms do not compromise data privacy. To mitigate privacy risks effectively, there is a call for the development of new methods to evaluate the loss of privacy based on training behaviour (Ding et al. 2022). Privacy breaches should be quantified, and subjective metrics assessing the seriousness of such breaches should be devised.

xAI models must explicitly consider privacy concerns throughout their lifecycle (Ali et al. 2023). Upholding the right to privacy is paramount, and xAI solutions should prioritise data integrity while processing data in a privacy-preserving manner. Protocols should dictate data access based on users and specific circumstances to maintain privacy. Srivastava et al. also remark that while fusion of outputs from xAI models can be powerful, it can also compromise overall model transparency and explainability. Moreover, if model explainability is reversible through knowledge fusion, it can undermine the model's differential privacy (Srivastava et al. 2022). Thus, it may so happen that explanations in xAI are viewed as potential privacy vulnerabilities (Nadeem et al. 2022). Generating privacy-preserving explanations is an emerging challenge, with limited existing work in this area (Nadeem et al. 2022). Balancing the need for transparency with privacy preservation is crucial to pre-

vent adversaries from exploiting explanations. Keeping this balance between the transparency and secrecy required to maintain user trust and protect confidential data is impossible without the application of privacy preservation techniques, such as federated learning. In addition, xAI plays a significant role in ensuring AI systems adhere to privacy policies and regulations, such as the EU GDPR (Javed et al. 2023).

Summing up, xAI has a dual impact on safety and security by enhancing error detection and prevention while raising privacy and security concerns (Hulsen 2023). Transparency in AI decisions can reveal sensitive information and potential vulnerabilities. Hence, careful consideration and risk mitigation measures are crucial, particularly in domains like healthcare where personal data protection is paramount. Combining algorithm explainability with the methods aimed at preserving privacy can contribute to striking the right balance (Hulsen 2023).

Potential opportunities The tension between xAI's drive for transparency and the need for privacy protection presents an opportunity for AI to innovate in developing privacy-preserving explanatory models that align with ethical standards and regulations like the GDPR. This challenge promotes the advancement of techniques such as federated learning, which could enable xAI systems to maintain algorithmic transparency without compromising data confidentiality, thus fostering trust and safety in AI applications, particularly in sensitive sectors such as healthcare.

3.5 Developing evaluation metrics

Numerous researchers notice that evaluation metrics form a critical aspect of xAI, directly impacting the assessment of xAI algorithms' ability to provide meaningful explanations (Pawlicka et al. 2024; Pawlicki et al. 2024). To establish a strong foundation for xAI, one or more evaluation metrics should facilitate a profound assessment of xAI algorithms' explanatory capabilities (Ding et al. 2022). Without such metrics, the literature remains weakened and lacks a consistent basis for evaluation. These metrics should quantitatively communicate how effectively an AI algorithm adheres to the facets of explainability.

The aforementioned absence of consensus on the definitions of interpretability and explainability poses challenges in metric development. The fact that there are no ground truth explanations and the cost associated with obtaining such explanations, coupled with the dynamic nature of optimal explanations, make metric development a complex task (Saeed and Omlin 2023).

Evaluation metrics for xAI predominantly fall into two categories: objective and human-centred evaluations (Saeed and Omlin 2023). Objective metrics are quantifiable mathematical measures, while human-centred evaluations rely on user studies. Further research is essential to establish objective metrics that can comprehensively assess xAI approaches across different contexts, models, and applications. A holistic evaluation approach integrates both human-centred and objective evaluations, aimed at comprehending how each metric contributes to the assessing of explanations (Saeed and Omlin 2023). Robust methodologies for human trials and evaluative measures of subjective explanations can aid in establishing agreed-upon criteria for human-centred evaluations.

Designing xAI for specific stakeholder categories, such as in IDS, necessitates customized evaluation metrics. At present, there is no unified agreement on the metrics for explanations, and finding the best explanation for each user type remains a challenge. Yet, as already

mentioned, explanations when misclassified due to attacks or poor data quality, can have severe consequences, such as significantly impacting Cyber Security Operation Centres (Neupane et al. 2022). Methods for auditing previously incorrect explanations, ideally performed by the explainable IDS itself, are crucial. Verification of explanations for correctness and robustness is also essential, especially when using xAI for model verification (Nadeem et al. 2022). In other words, ensuring that explanations themselves are accurate and reliable is a critical aspect of evaluation.

In addition to this, measuring explainability presents challenges due to its subjective nature (Hulsen 2023). Unlike accuracy and performance metrics, the quality of an explanation can vary substantially. Researchers have attempted to develop assessment methods for explainability, but achieving a universally accepted metric remains challenging.

Potential opportunities Developing evaluation metrics for xAI is challenging due to the subjective nature of explainability and the absence of standardized definitions. Metrics must balance objective quantification and human-centred assessments to ensure AI explanations are both accurate and user-friendly. Yet, this issue creates multiple opportunities. These include standardizing metrics to provide reliable benchmarks, innovating in metric development to accommodate dynamic and context-specific explanations, and enhancing AI's trustworthiness in critical applications. Additionally, these challenges encourage collaborative cross-disciplinary research and offer commercial advantages for companies seeking to market transparent AI products. Lastly, they provide educational opportunities to train AI practitioners in new methodologies, advancing the integration of AI into society with better understanding and accountability.

3.6 Increasing complexity, systematic instability and multiplicity of explanations

Researchers also point to the challenge of systematic instability in the explainability of machine learning.

The early neural networks, characterized by a single layer, were quite straightforward to grasp. However, the evolution of deep learning, accompanied by the development of complex algorithms like Deep Belief Networks (DBNs) and Generative Adversarial Networks (GANs), driven by increasing computational power, has led to a significant rise in algorithmic complexity (Hulsen 2023). As ML algorithms have become increasingly complex, the task of making them explainable becomes more daunting. The intricate nature of these algorithms can result in systematic instability, where multiple valid approaches can be generated for the same input and prediction objectives. This complexity poses challenges for automated explanations, given that the details of the explanations might vary across several valid models (Ding et al. 2022).

It is anticipated that AI algorithms will become even more complex in the future. While this increasing complexity presents challenges for explainability, ongoing research in the field of explainable artificial intelligence may lead to the development of new techniques that make it easier to comprehend and explain complex AI models (Hulsen 2023).

Potential opportunities The complexity of developing evaluation metrics for xAI underscores the opportunity to pioneer robust, universally accepted quantitative and human-centred benchmarks that accurately reflect an AI system's explanatory capabilities. This pursuit could lead to innovative approaches in metric design, fostering a standardized assessment of AI explanations across various domains, and ensuring that explanations not only enhance

user understanding but also maintain accuracy and reliability, particularly in sensitive applications like intrusion detection systems.

3.7 Adversarial use of xAI

As already established, in the rapidly evolving digital world, systems and software are invariably susceptible to an array of security vulnerabilities and cyber intrusions. AI algorithms are frequently dubbed as one of the best solutions for detecting these threats (Pawlicki et al. 2022, 2023). Nonetheless, the majority of AI-driven security measures produce 'black-box' classification results, which can be inscrutable even to seasoned security experts. Therefore, progressing towards explainable security solutions emerges as a prospective avenue to enhance the protection of both public and proprietary systems. (Ding et al. 2022). Yet, researchers notice that while xAI holds promise for improving cybersecurity, it also introduces an array of new vulnerabilities.

The researchers point it out that, of late, adversaries have leveraged various techniques to exploit the xAI components embedded within cybersecurity frameworks.

- Poisoning attacks have been deployed to tamper with or impair the training datasets.
- Adversarial attacks were utilized to circumvent verification processes, including those reliant on xAI-driven facial recognition systems.
- Counterfactual gaming attacks were employed to tweak defence and offensive protocols, capitalizing on the susceptibilities inherent in xAI-integrated cybersecurity structures (Srivastava et al. 2022).
- Confidentiality attacks: Explanations provide supplementary insights into the internal mechanics of an operational model, simplifying the task of recreating both the model and its training data (Nadeem et al. 2022).
- Model inversion attacks provide adversaries with the capability to reverse-engineer training data based on model outputs. With the aid of explanations, these adversaries can replicate the model with better precision. For instance, xAI-conscious model inversion tactics can proficiently retrieve images originating from the training dataset (Nadeem et al. 2022).
- Membership inference attacks operate on the premise that an adversary possesses certain inputs and seeks to ascertain if these were incorporated during the training phase (Nadeem et al. 2022).
- Model extraction attacks have facilitated adversaries in deducing both the architecture and parameters of a model based on its predictions. In this case, explanations provide extra information for adversaries concerning which attributes to modify in order to disrupt the model's appropriate operation, either during deployment (evasion attacks) or during training (poisoning and backdoor attacks) (Nadeem et al. 2022).On top of that, Srivasta and colleagues postulate that to maintain the balance between security and user-friendliness, cybersecurity professionals must integrate both security and usability considerations, when conceptualizing and designing new antivirus solutions to ensure they are user-centric. Tools to analyse and debug the model and its data internals should be introduced to safeguard the integrity and security of emergent cybersecurity structures against potential adversarial actions (Srivastava et al. 2022).

The extensive body of research on adversarial learning clearly indicates that sole dependence on performance indicators is a dangerous approach, given that xAI introduces an added avenue of assault for adversaries in the domain of cybersecurity. Both retrospective explanation tools and inherently interpretable models exhibit vulnerability to slight adversarial disturbances. This introduces a novel spectrum of attacks wherein creators of malicious software can evade detection and concurrently obfuscate the characteristics employed for such evasion. Consequently, it becomes crucial for model architects to fortify explanation mechanisms against adversarial disruptions, as underscored by (Nadeem et al. 2022).

From the vantage point of proactive security measures, xAI may also provide decision support to adversaries for better attack formulation (Nadeem et al. 2022). To address these concerns, it is recommended to either use a robust explainer or conduct explanation verification under adversarial settings (Nadeem et al. 2022).

Potential opportunities The emergence of adversarial exploitation of xAI systems presents a significant opportunity to fortify AI against such vulnerabilities. By understanding how adversaries might manipulate explanations, researchers and practitioners can develop more resilient AI models. This includes devising methods for safeguarding explanation mechanisms and integrating security within the design of user-friendly, explainable cybersecurity tools, thus enhancing overall system robustness against sophisticated cyber threats.

3.8 User-centric validation and evaluation

This challenge again relates to the importance of validation and evaluation of xAI; precisely, a number of researchers believe that they need to be as user-centric as possible. The challenge highlights several key elements that emphasise the need to involve end-users in the validation and evaluation processes of AI systems while tailoring explanations to meet their needs and expectations (Ali et al. 2023).

It has been emphasised that for ensuring transparency, accountability, and fairness, AI systems ought to undergo validation with real-world users. Given AI's increasing role in automated decision-making systems that often involve personal information, user validation becomes crucial. This aligns with regulations such as the General Data Protection Regulation (GDPR), which acknowledges the significance of individual rights in AI systems.

In line with this, (Ali et al. 2023) highlight the importance of sociological studies in understanding how individuals react to explanations. This includes psychological research, which helps determine what people perceive as a good explanation. Also, qualities like conciseness, social acceptability, and contrastiveness are essential in making explanations effective and acceptable to users. Lastly, (Doshi-Velez and Kim 2017) suggest a multi-layered approach to evaluating explainability. This approach involves conducting experiments with users engaged in genuine real-world tasks, executing foundational experimental activities with individuals, and employing surrogate tasks and benchmarks that have been corroborated by prior studies. It emphasises the need for a comprehensive evaluation framework.

It must be noted that despite these recommendations, there is a notable gap in user-based studies within the field. Few studies have addressed user experiments, leaving room for more comprehensive coverage to address all the aspects of user interaction with AI systems. The importance of interdisciplinary cooperation and user-centred approaches is also stressed, as this can help bridge the gaps between xAI research and practical applications.

Furthermore, it is vital to recognize that humans should not be replaced by AI algorithms, especially in domains like healthcare. While AI can excel in specific tasks, humans bring proficiency in the domain, extensive experiential knowledge, and adeptness in innovative thought processes. The combination of AI and human expertise, a.k.a. 'AI-assisted decision making', often yields the most desirable results. This hybrid approach not only enhances accuracy but also makes decisions more explainable, as experts can integrate their domain knowledge with the insights provided by AI (Hulsen 2023).

Potential opportunities The challenge of ensuring user-centric validation and evaluation in xAI underscores the opportunity to enhance AI systems through user engagement, aligning with transparency, accountability, and fairness principles. This focus could lead to the development of new methodologies that incorporate real-world user feedback and sociological insights, creating AI systems that are more attuned to individual needs and societal values. Moreover, the necessity of integrating human expertise with AI, especially in sensitive areas like cybersecurity, highlights the potential for AI-assisted decision-making that benefits from the synergistic blend of human intuition and AI precision.

3.9 Usability and interpretability

The previous challenge is closely related to another one mentioned by multiple researchers, i.e., the challenges related to the usability and interpretability of the xAI systems. It involves issues such as the simplicity of explanations, the usability of explanation methods, and the propensity of users to make biased inferences based on xAI explanations.

It is worth recalling the study by Evans et al., in which the participants clearly underscored the importance of simplicity in explanation methods. In addition, according to the study, the participants highly valued straightforward explanations, and complexity in interface elements that necessitate additional scrutiny or training for interpretation may counteract the desired transparency. Furthermore, this predilection for simplicity extends to the preference for visual explanations. However, it is noteworthy that introducing various modalities into primarily visual workflows can impose an undesirable cognitive burden on users (Evans et al. 2022).

To back this up, in the study by Evans et al., a wide range of modalities emerged as valuable sanity checks, flagging instances where AI outputs exhibited clear inaccuracies (Evans et al. 2022). These findings suggest that explanations can serve as powerful tools to discredit AI outputs in specific situations. The study also revealed an inclination towards affirmatory confirmation bias in the context of explanations that aligned closely with or did not significantly challenge human expectations. Participants often sought relatable causal reasoning within the explanations to better comprehend the AI's decision-making processes. Additionally, the trial period emerged as an important ingredient in building users' trust in AI systems. It played a fundamental role in establishing the credibility of AI-powered assessments (Evans et al. 2022).

This divergence of attitudes towards human versus machine fallibility was a noteworthy observation. According to Evans et al., in their study, the AI-sceptical participants tended to reject the system when deviations between human and AI assessments occurred. Conversely, AI-optimistic individuals believed conflicting or even incorrect AI results were of value, serving as a means to remain vigilant. In these cases, the objective stance of AI, unaf-

fected by elements such as the participant's tenure or expertise, was deemed a commendable quality, often absent in consecutive opinions from human colleagues (Evans et al. 2022).

Potential opportunities The challenges of usability and interpretability in xAI underscore the opportunity for more intuitive, user-friendly AI systems that prioritise simple and clear explanations to build user trust and mitigate biases, fostering a productive human-AI partnership.

3.10 Interdisciplinary collaboration for advancing xAI

In the field of xAI, there is a strong emphasis on fostering interdisciplinary collaboration. This highlights the crucial role of researchers from different fields, including psychology, behavioural and social sciences, human-computer interaction (HCI), physics, and neuroscience, coming together to advance human-centred AI and expand the use of xAI in critical applications. This collaborative approach opens up avenues for multidisciplinary research, exploring how insights from these diverse fields can enhance xAI methods. It also underscores the importance of conducting user experiments and involving humans in the development and utilization of xAI models. Additionally, this approach stresses the benefits of integrating insights from philosophy, cognitive psychology/science, and social psychology into xAI research.

In alignment with the human-computer interaction community's perspective, the authors, including Antoniadi et al., emphasise the need for interdisciplinary cooperation and user-centred approaches to explainability. They also highlight the significance of understanding the motivations behind explainability to choose the right xAI techniques, anticipate their limitations, and enhance the user experience of xAI design (Antoniadi et al. 2021). xAI is envisioned as interactive and human-like, providing tailored explanations for different users. However, challenges arise due to the current limitations of available xAI techniques and competing goals.

From a technological standpoint, enabling technologies and methods for xAI can be categorized into four fundamental research areas:

1. Data Science: AI and machine learning algorithms are data-hungry, requiring substantial data for better predictions and decisions. This also applies to generating better explanations and justifications, making data science a crucial element in the explainability process.
2. Artificial Intelligence/Machine Learning: Generating explanations requires a computational process, and the authors propose using AI/ML as a computational approach to explain AI/ML, which presents an intriguing research path.
3. Human Science: To create artificial explanations that resonate with human understanding, it is important to study how people explain their decisions and behaviour to one another. Drawing insights from the humanistic disciplines can pave the way for pioneering explainable models.
4. Human-Computer Interaction (HCI): The user's comprehension and trust in AI systems depend in part on how they interact with the technology. Given HCI's focus on understanding and empowering users, techniques from this field can contribute to the development of transparent and user-friendly systems (Adadi and Berrada 2018).

Potential opportunities The call for interdisciplinary collaboration in xAI underscores the potential for breakthroughs in human-centred AI by integrating diverse fields like psychology, neuroscience, data science, and human-computer interaction (HCI). This approach not only enriches xAI methods with varied insights but also emphasises the importance of user involvement in both the development and application of xAI models. Through this multidisciplinary lens, the pursuit of explainability in AI becomes a confluence of technological advancement and humanistic understanding, aimed at creating AI systems that are both transparent and resonant with human users.

3.11 Time-sensitive explanations

Another significant challenge indicated by the researchers in the domain of xAI is the temporal aspect of explanations. This challenge revolves around the necessity to adapt the process of generating explanations to accommodate varying time constraints and meet the diverse requirements of users. This encompasses situations where explanations must be swiftly generated, enabling users to make rapid decisions, like in the case of the application of xAI in cybersecurity, as well as scenarios in which users have the luxury of dedicating more time to comprehending complicated explanations. Striking the right balance between the efficiency of explanation generation, the available computing resources, and the unique needs of end-users emerges as a key aspect of this challenge (Saeed and Omlin 2023).

Furthermore, when it comes to practical applications of xAI in the field of security, it is imperative to address the time complexity of the underlying algorithms (Srinivasu et al. 2022). An additional challenge surfaces in the form of reconciling the provision of timely responses to user queries with the maintenance of computational efficiency. Interactive explanation systems must swiftly respond to user interactions to ensure a seamless and engaging user experience. Nevertheless, the task of generating precise and informative explanations within a limited response time window can be arduous, necessitating trade-offs that are contingent upon the specific domain and the computational resources at hand (Yan et al. 2022).

To address these challenges effectively, one key criterion is either the real-time delivery of explanations, or providing explanations with minimal delay. This real-time aspect is vital to support timely decision-making, and it implies a temporal scale on the order of seconds rather than tens of minutes (Swamy et al. 2023).

Potential opportunities The challenge of providing time-sensitive explanations in xAI highlights the need for adaptable systems capable of generating explanations with varying levels of depth according to time constraints, particularly in high-stakes environments like cybersecurity. This calls for innovative solutions that balance computational efficiency with the immediacy of user needs, ensuring that AI systems can offer real-time or near-real-time insights to support swift and informed decision-making.

3.12 Ensuring fairness, accountability, and trustworthiness

This challenge accentuates the imperative for xAI to actively promote fairness by addressing inherent biases within machine learning algorithms and by providing transparency in the decision-making processes. It also emphasises the significance of accountability, wherein

a designated entity bears responsibility for AI decisions with significant consequences, and delineates how xAI can contribute to reinforcing accountability systems.

The researchers underline the unceasing necessity of upholding fairness, accountability, and transparency throughout the design of xAI techniques and once again highlight the critical role of quantitative metrics in objectively assessing the quality of explanations (Saeed and Omlin 2023; Ali et al. 2023; Nadeem et al. 2022).

As pointed out by Nadeem and colleagues, one can even derive equitable explanations from inherently biased models. In settings with stringent regulations mandating companies to furnish explanations alongside their opaque models, there is potential for such explanations to be misused in 'fairwashing' endeavours—projecting a deceptive image of model fairness when it is inherently skewed. Hence, it is advisable to lean towards intrinsically interpretable models. If this is not feasible, it becomes critical to establish a congruent relationship between a model and its respective explanation, e.g., by training a surrogate model that is certified as equivalent (Nadeem et al. 2022).

Potential opportunities The imperative to ensure fairness, accountability, and trustworthiness in xAI underscores the opportunity for AI to evolve with ethical considerations at its core. It calls for innovative research and design approaches that inherently integrate these values into AI systems, using quantitative metrics for objective assessment. This challenge also brings to light the potential for 'fairwashing' and the necessity of developing intrinsically interpretable models, or at the very least, tightly coupled models and explanations that reflect true fairness, thereby fostering genuine trust in AI applications.

3.13 Quality natural language explanations

Another challenge that those who adopt xAI may face is ensuring the quality and clarity of natural language explanations. The primary objective is to guarantee the precision and accessibility of these explanations to users. Several significant sub-challenges have been identified, including the Evaluation Challenge, which involves developing robust methods for evaluating explanation quality in terms of the language; the Vague Language Challenge, which requires a delicate balance between enhancing human understanding and averting misinterpretation; the Narrative Challenge, aimed at rendering symbolic reasoning more user-friendly; the Communicating Data Quality Challenge, focusing on addressing data quality concerns; and the Coherent Report Generation Challenge in the Medical Domain, a specific hurdle in generating coherent textual reports akin to human doctors' notes. These challenges collectively underscore the importance of not solely generating explanations within xAI but also ensuring their quality, clarity, and relevance within the user's context (Saeed and Omlin 2023; Islam et al. 2022).

Potential opportunities The challenge of crafting high-quality natural language explanations in xAI emphasises the need for AI to communicate complex concepts in an understandable and precise manner, balancing clarity with accuracy to enhance user trust and the applicability of AI insights. Still, by addressing these challenges, there is a significant potential to enhance user trust and the practical usability of AI systems. These efforts could dramatically improve how explanations are crafted and perceived, making xAI more effective across various domains.

3.14 Data quality

This challenge concerns the quality of data used in AI system development. In their work, Kalyanathaya et al. aptly note that the 'varied quality of data in each decision-making process may impact the outcome' (Kalyanathaya 2022). Additionally, Korica emphasises the necessity of auditing data quality to ensure that machine learning systems are trained on high-quality, unbiased, and accurate data (Korica et al. 2021). The provision of explanations in AI systems relies heavily on the quality of the underlying data. Data bias, incompleteness, and inaccuracies can significantly affect the quality of explanations (Saeed and Omlin 2023). Training AI systems with subpar data may lead to reduced accuracy, unfair decisions, and diminished explainability. To address this, it is crucial to be conscious of data collection methods and transparently communicate any data-related limitations to maintain the accuracy of explanations (Saeed and Omlin 2023).

Also in Saeed and Omlin (2023), the authors discuss the challenges of producing explanations for AI models employing imputation of missing data. They recommend the use of disclaimers alongside explanations to educate end-users about potential risks associated with incorrect explanations. Additionally, they also suggest that future studies should focus on developing practical and measurable methods to communicate data quality to users, potentially by proposing dimensions for data quality assessment (Saeed and Omlin 2023).

Potential opportunities The challenge of data quality in AI underscores the need for rigorous data auditing to ensure that AI systems are trained on data that is unbiased, accurate, and of high quality. This directly affects the reliability of AI explanations, necessitating transparent communication of data limitations and the development of methods to assess and convey data quality to users. Addressing this challenge is critical for maintaining the integrity and explainability of AI decisions.

3.15 Explainability for diverse data types

It has also been remarked that while xAI research has predominantly concentrated on image and text data, other data types have received comparatively less attention. These overlooked data categories encompass a wide spectrum, including time-series, graphs, spatio-temporal data, and various structured data formats.

One potential avenue to address this challenge involves the transformation of non-image data into representations akin to images using visualization techniques. This transformation creates opportunities to unveil explanations by highlighting salient pixels and features. Yet, relying solely on this method may prove insufficient due to the inherent disparities between various data types. For instance, explanation approaches originally devised for image or text data may require adaptation to effectively interpret graph data. Interpretations that are readily comprehensible in the context of image explanation approaches, such as saliency maps, may demand expert knowledge when applied to time-series data. Moreover, there is a pressing need to develop novel approaches tailored explicitly to explain the unique characteristics of non-image or non-text data types. This entails embracing the structural intricacies within graph data and effectively handling multivariate time series data with varying temporal lengths. In today's era of AI systems harnessing a diverse array of data types, it is imperative that explainability approaches exhibit adaptability to accommodate this heterogeneity (Saeed and Omlin 2023).

Potential opportunities The challenge of extending explainability to diverse data types in xAI emphasises the need for innovative methods tailored to complex formats like time-series and graphs, ensuring intuitive and accurate interpretations across all kinds of AI applications.

3.16 Information gap and confirmation bias

This category highlights challenges associated with information gaps in explanations provided by explainable artificial intelligence systems and the potential for confirmation bias in user interpretations.

Research evidence suggests that each omitted detail in an explanation, whether for clarity or due to intrinsic constraints, creates room for user speculation. Essentially, any voids or vaguenesses in the disseminated information may be interpreted or 'completed' based on the user's individual anticipations and comprehension. Such phenomena introduce the potential for confirmation bias, especially in explanations that do not comprehensively represent the crucial determinants of a model's output. This could, consequently, lead users to harbour a false sense of confidence in the pertinent AI system (Evans et al. 2022).

The issue is notably pronounced when considering user inclinations towards explanations deemed 'intuitively understandable' and elements of explainability, since these naturally leverage pre-existing notions held by the user. Given that these prior beliefs differ from one individual to another and across scenarios, there exists an inherent ambiguity regarding the precise information users perceive from such explanations. It is important to emphasise that even components not intentionally crafted as explanations might still fall under this observation, as discussed by Evans et al. (2022).

This underscores the imperative for xAI methodologies that provide unambiguous information to the user. An ideal trust score aligned with this principle could amalgamate various determinants, accessible upon request, with each constituent possessing a clear and interpretable significance. For instance, factors might indicate that the data deviates from the model's typical distribution, the image resolution is suboptimal, or the presence of contradictory features has been detected, among other considerations (Evans et al. 2022).

Potential opportunities The challenge posed by information gaps and confirmation bias in xAI calls for strategies to deliver clear and complete explanations that mitigate the user's tendency to fill in blanks with their assumptions. This requires xAI to be designed with much attention to detail, ensuring that users receive all the necessary information to make informed decisions without defaulting to their biases. Developing comprehensive trust scores that transparently communicate the reliability of AI predictions and the quality of underlying data could be key in addressing this issue.

3.17 The less studied challenges in explainable artificial intelligence

As presented, in the realm of explainability applied in cybersecurity, research has made significant strides in addressing prominent challenges, such as interpretability, transparency, security, privacy, and the temporal aspects of explanations. However, the path to enhancing the efficacy of xAI systems is not solely illuminated by well-trodden avenues. In this subsection, the less-frequented but equally crucial challenges within xAI will be mentioned; the issues that have garnered fewer mentions in the literature but possess the potential to

reshape the landscape of explainability and accountability in AI systems. These potentially overlooked challenges, while less conspicuous, demand the community's attention and exploration as they may hold the key to unlocking the full potential of xAI in an ever-evolving technological landscape.

Some of such challenges revolve around the evaluation and optimization of xAI visualizations and interpretability methods. These challenges include the reluctance of human attention to effectively utilize xAI explanation maps, the absence of quantifiable metrics for assessing explanatory map precision and completeness, and the need to reevaluate the utility of visualization approaches in mission-critical applications. Furthermore, the researchers emphasise the significance of diverse explanation types and the role of human-generated data in enhancing explanatory maps (Ding et al. 2022; Islam et al. 2022; Javed et al. 2023).

Then, another challenge is related to to ensuring fidelity and causality in explanations. This challenge emphasises the importance of context-aware intelligent systems employing complex AI solutions, which can be challenging for human comprehension. The absence of such systems risks client dissatisfaction and undermines the qualification of xAI for critical applications. Still, the automatic generation of causative explanations offers promise in mitigating this challenge (Ding et al. 2022).

Another critical challenge centres on Causal Interpretability and Explanation, advocating the provision of insights into why AI algorithms make predictions, rather than solely how. Striking a balance between predictive efficacy and causal reasoning, particularly when confounding variables are at play, warrants extended exploration. Moreover, devising dependable techniques for the assessment and validation of causal interpretability stands as a paramount research avenue within the xAI realm (Saeed and Omlin 2023).

In the literature, one challenge that stands out is maintaining accurate representations of AI capabilities and limitations. This challenge centres on the delicate balance between providing relatable explanations that align with user understanding and the risk of implying human-like reasoning in AI systems, which fundamentally differ in their processing of information from humans. As put by Evans, ensuring explanations 'speak the language of the user' may inadvertently misrepresent AI capabilities and boundaries. This underscores the pressing call for investigations into user-centric xAI methodologies that provide a window into the inner mechanisms of opaque systems, yet respect and reflect their inherent abstract and statistical essence. Developing such tools must go hand-in-hand with educating users about AI capabilities and constraints. Achieving this balance should be a core principle in xAI design, supplemented by active user outreach and training on the fundamentals and current state of machine learning and AI applications in their respective domains (Evans et al. 2022).

Another noteworthy challenge that Evans brings attention to is balancing the benefits and risks of incorporating explainability into AI systems, especially in sensitive domains like medicine. The growing pressure to enhance explainability introduces the risk of applying explanation-generating methods without fully considering potential second-order effects. To address this, AI developers must carefully assess the advantages and drawbacks of including explainability elements, engaging in a tight feedback loop with potential users while considering the lenses of social and cognitive theories of Human-Computer Interaction (Evans et al. 2022). Should the practicality and user-friendliness of explainability components fail to harmonize with the prospective biases they could instigate, it may indicate their inappropriateness for the use case. In such instances, it becomes imperative to either re-evaluate the

AI system at a fundamental level or, if necessary, forgo explainability altogether to avoid the risk of misleading users (Evans et al. 2022).

In their work, Saeed and Omlin discuss a spectrum of challenges related to the model and method development. Scalability, as the first challenge, becomes prominent when dealing with a multitude of cases requiring predictions and explanations. Methods like Local Interpretable Model-agnostic Explanations (LIME) are powerful but computationally expensive for large datasets. The Feature Dependence Challenge arises from correlated features, making it difficult to attribute importance accurately, especially in sensitivity analyses. Existing Explanation Techniques, represented by heatmaps, although valuable, lack clarity regarding feature-prediction relationships, necessitating more abstract meta-explanations. Model-based and post-hoc explainability confronts issues of prediction accuracy and feature engineering tools, requiring improvements for interpretable yet precise models. Post-hoc methods must also tackle the challenge of describing model behaviour adequately. Perturbation-Based Methods grapple with determining optimal input perturbations due to combinatorial complexity, demanding empirical studies for cross-domain applications (Saeed and Omlin 2023).

The remaining identified challenges are as follows:

- adapting xAI methods for tabular data, emphasising the unique characteristics of tabular information that distinguish it from visual or textual data, including feature interdependencies, constant attributes, and chronological aspects. It underscores the imperative to develop distinct methods tailored for interpreting tabular data, recognizing that the existing categories of xAI methods may not seamlessly apply to this domain (Ding et al. 2022).
- quantification and communication of uncertainties within xAI systems. This encompasses measuring and expressing uncertainties inherent in machine learning models, as well as those introduced during model refinement. The objective is to offer users reliable information about uncertainties within AI models and their explanations, emphasising the importance of rigorous quantification and communication at various phases of the model's lifecycle (Saeed and Omlin 2023).
- the economic perspective of xAI, potential business value, but also acknowledging the associated costs and challenges. These include explanation design, audit log creation, trade secrets protection, innovation speed, decisional flexibility, security, and privacy considerations. The importance of accurate cost estimation and economic analysis is emphasised to make informed decisions regarding the adoption of xAI in business contexts (Saeed and Omlin 2023).

Potential opportunities The less studied challenges in xAI, particularly in the context of cybersecurity, highlight the potential for groundbreaking research in areas like optimizing xAI visualizations, ensuring fidelity and causality in explanations, and maintaining accurate representations of AI capabilities. These underexplored areas present opportunities to refine xAI utility in mission-critical applications, to enhance the balance between predictive efficacy and causal reasoning, and to develop user-centric methodologies that accurately convey AI's abstract nature. Addressing these challenges requires a concerted effort to create scalable, accurate, and user-friendly explanation systems that are sensi-

tive to the subtleties of human-AI interaction and the complexities of underlying model behaviours.

4 Discussion and conclusions

This paper has gathered and discussed 25 various challenges to the adoption of explainability in intrusion detection. They have been presented in Fig. 4. In this figure, the dual nature of the challenges identified in the study has been illustrated. Specifically, the figure demonstrates that each of the 25 challenges listed in Table 2 can also be viewed as potential opportunities. The authors of this study believe that this perspective embraces the dynamic and evolving nature of the cybersecurity landscape, where obstacles encountered by organisations and researchers often present opportunities for innovation, improvement, and strategic advantage. In turn, when viewing challenges as opportunities, one can shift their focus from simply mitigating risks to leveraging these challenges to develop more adaptive solutions.

A careful examination of the identified diverse challenges let the authors emphasise the pressing need for continued research and development in the field of explainable intrusion detection, as achieving transparency and interpretability in these systems is paramount for ensuring the security and trustworthiness of digital infrastructure and ultimately enhancing the resilience of the cybersecurity defences in an increasingly complex threat landscape.

It must once again be underlined that the 'challenges' presented in this paper do not necessarily mean 'problems'. In the ever-evolving world of AI, what one may perceive as an impediment, another might see as an avenue for innovation. Most of these challenges can be seen as 'opportunities', or 'directions' guiding further research and development. In fact, the analysis of the gathered material let the authors conclude that in the subject literature, the same issue often is presented either as a challenge or an opportunity, depending on the researcher's outlook on the matter, and their way of reasoning.

This duality of challenges as opportunities underscores the dynamic nature of the field of xAI. Rather than simply hurdles to overcome, these challenges can serve as catalysts, driving innovation and progress. Each obstacle may be reframed as a stepping stone towards a more robust, adaptable, and efficient intrusion detection ecosystem. Adopting this perspective encourages a collaborative and forward-thinking approach, turning potential setbacks into breakthrough moments. It makes it possible for the research community to harness the potential within these challenges and positively influence the general acceptance of AI.

Acknowledgements This work is partially funded under the *HE AI4CYBER* project, with the support of the European Commission and the Horizon Europe Program, under Grant Agreement No. *101070450*. This work is partially funded under the *HE ULTIMATE* project, with the support of the European Commission and the Horizon Europe Program, under Grant Agreement No. *101070162*.

Author contributions MP: writing—original draft, methodology, investigation, formal analysis, conceptualization. AP: writing—original draft, visualization, methodology, investigation, formal analysis, conceptualization. RK: methodology, investigation. MC: conceptualization, formal analysis, funding acquisition, supervision, validation, writing—original draft.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Adadi A, Berrada M (2018) Peeking Inside the black-box: a survey on explainable artificial intelligence (XAI). IEEE Access 6:52138–52160. https://doi.org/10.1109/ACCESS.2018.2870052

Ali S, Abuhmed T, El-Sappagh S, Muhammad K, Alonso-Moral JM, Confalonieri R, Guidotti R, Del Ser J, Díaz-Rodríguez N, Herrera F (2023) Explainable artificial intelligence (XAI): what we know and what is left to attain trustworthy artificial intelligence. Inf Fusion. https://doi.org/10.1016/j.inffus.2023.101805

Amirian S, Carlson LA, Gong MF, Lohse I, Weiss KR, Plate JF, Tafti AP (2023) Explainable AI in orthopedics: challenges, opportunities, and prospects. arXiv:2308.04696

Antoniadi AM, Du Y, Guendouz Y, Wei L, Mazo C, Becker BA, Mooney C (2021) Current challenges and future opportunities for XAI in machine learning-based clinical decision support systems: a systematic review. Appl Sci 11(11):5088. https://doi.org/10.3390/app11115088

Brik B, Chergui H, Zanzi L, Devoti F, Ksentini A, Siddiqui MS, Costa-Pérez X, Verikoukis C (2023) A survey on explainable AI for 6G O-RAN: architecture, use cases, challenges and research directions. arXiv:2307.00319

Capuano N, Fenza G, Loia V, Stanzione C (2022) Explainable artificial intelligence in cybersecurity: a survey. IEEE Access 10:93575–93600. https://doi.org/10.1109/ACCESS.2022.3204171

Choraś M, Pawlicki M, Puchalski D, Kozik R (2020) Machine learning—the results are not the only thing that matters! What about security, explainability and fairness? ICCS 4:615–628. https://doi.org/10.1007/978-3-030-50423-6_46

Ding W, Abdel-Basset M, Hawash H, Ali AM (2022) Explainability of artificial intelligence methods, applications and challenges: a comprehensive survey. Inf Sci 615:238–292. https://doi.org/10.1016/j.ins.2022.10.013

Doshi-Velez F, Kim B (2017) Towards a rigorous science of interpretable machine learning. arXiv:1702.08608

Eldawlatly A, Alshehri H, Alqahtani A, Ahmad A, Al-Dammas F, Marzouk A (2018) Appearance of population, intervention, comparison, and outcome as research question in the title of articles of three different anesthesia journals: a pilot study. Saudi J Anaesth 12(2):283. https://doi.org/10.4103/sja.SJA_767_17

Evans T, Retzlaff CO, Geißler C, Kargl M, Plass M, Müller H, Kiehl T-R, Zerbe N, Holzinger A (2022) The explainability paradox: challenges for xAI in digital pathology. Future Gener Comput Syst 133:281–296. https://doi.org/10.1016/j.future.2022.03.009

Goodman B, Flaxman S (2017) European union regulations on algorithmic decision making and a "right to explanation". AI Mag 38(3):50–57. https://doi.org/10.1609/aimag.v38i3.2741

Haque AB, Islam AKMN, Mikalef P (2023) Explainable artificial intelligence (XAI) from a user perspective: a synthesis of prior literature and problematizing avenues for future research. Technol Forecast Soc Change 186:122120. https://doi.org/10.1016/j.techfore.2022.122120

Hulsen T (2023) Explainable artificial intelligence (XAI) concepts and challenges in healthcare. AI 4(3):652–666. https://doi.org/10.3390/ai4030034

Islam MR, Ahmed MU, Barua S, Begum S (2022) A systematic review of explainable artificial intelligence in terms of different application domains and tasks. Appl Sci 12(3):1353. https://doi.org/10.3390/app12031353

Jagatheesaperumal SK, Pham Q-V, Ruby R, Yang Z, Xu C, Zhang Z (2022) Explainable AI over the internet of things (IoT): overview, state-of-the-art and future directions. IEEE Open J Commun Soc 3:2106–2136. https://doi.org/10.1109/OJCOMS.2022.3215676

Javed AR, Ahmed W, Pandya S, Maddikunta PKR, Alazab M, Gadekallu TR (2023) A survey of explainable artificial intelligence for smart cities. Electronics 12(4):1020. https://doi.org/10.3390/electronics12041020

Kabir MH, Hasan KF, Hasan MK, Ansari K (2021) Explainable artificial intelligence for smart city application: a secure and trusted platform. arXiv:2111.00601

Kalyanathaya KP (2022) A literature review and research agenda on explainable artificial intelligence (XAI). Int J Appl Eng Manage Lett 6(1):43–59. https://doi.org/10.47992/IJAEML.2581.7000.0119

Korica P, Gayar NE, Pang W (2021) Explainable artificial intelligence in healthcare: opportunities, gaps and challenges and a novel way to look at the problem space, pp 333–342. https://doi.org/10.1007/978-3-030-91608-4_33

Lipton ZC (2016) The mythos of model interpretability. arXiv:1606.03490

Lundberg SM, Lee S-I (2017) A unified approach to interpreting model predictions. In: Guyon I, Luxburg UV, Bengio S, Wallach H, Fergus R, Vishwanathan S, Garnett R (eds) Advances in neural information processing systems, vol 30. Curran Associates, Long Beach, CA. https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf. Accessed 25 June 2024

Machlev R, Heistrene L, Perl M, Levy KY, Belikov J, Mannor S, Levron Y (2022) Explainable artificial intelligence (XAI) techniques for energy and power systems: review, challenges and opportunities. Energy AI 9:100169. https://doi.org/10.1016/j.egyai.2022.100169

Mathew A (2023) Explainable AI for intelligence analysis. Int J Eng Res Technol (IJERT) 12(2). https://www.ijert.org/explainable-ai-for-intelligence-analysis

Meske C, Bunde E, Schneider J, Gersch M (2022) Explainable artificial intelligence: objectives, stakeholders, and future research opportunities. Inf Syst Manage 39(1):53–63. https://doi.org/10.1080/10580530.2020.1849465

Mijwil M, Salem IE, Ismaeel MM (2023) The significance of machine learning and deep learning techniques in cybersecurity: a comprehensive review. Iraqi J Comput Sci Math. https://doi.org/10.52866/ijcsm.2023.01.01.008

Nadeem A, Vos D, Cao C, Pajola L, Dieck S, Baumgartner R, Verwer S (2022) SoK: explainable machine learning for computer security applications. arXiv:2208.10605

Neupane S, Ables J, Anderson W, Mittal S, Rahimi S, Banicescu I, Seale M (2022) Explainable intrusion detection systems (X-IDS): a survey of current methods, challenges, and opportunities. arXiv:2207.06236

Nwakanma CI, Ahakonye LAC, Njoku JN, Odirichukwu JC, Okolie SA, Uzondu C, Ndubuisi Nweke CC, Kim D-S (2023) Explainable artificial intelligence (XAI) for intrusion detection and mitigation in intelligent connected vehicles: a review. Appl Sci 13(3):1252. https://doi.org/10.3390/app13031252

Paredes JN, Teze JCL, Simari GI, Martinez MV (2021) On the importance of domain-specific explanations in AI-based cybersecurity systems (technical report). arXiv:2108.02006

Pawlicka A, Jaroszewska-Choras D, Choras M, Pawlicki M (2020) Guidelines for stego/malware detection tools: achieving GDPR compliance. IEEE Technol Soc Mag 39(4):60–70. https://doi.org/10.1109/MTS.2020.3031848

Pawlicka A, Pawlicki M, Kozik R, Kurek W, Choraś M (2024) How explainable is explainability? Towards better metrics for explainable AI, pp 685–695. https://doi.org/10.1007/978-3-031-44721-1_52

Pawlicki M, Kozik R, Choraś M (2022) A survey on neural networks for (cyber-) security and (cyber-) security of neural networks. Neurocomputing 500:1075–1087. https://doi.org/10.1016/j.neucom.2022.06.002

Pawlicki M, Pawlicka A, Kozik R, Choraś M (2023) The survey and meta-analysis of the attacks, transgressions, countermeasures and security aspects common to the Cloud, Edge and IoT. Neurocomputing 551:126533. https://doi.org/10.1016/j.neucom.2023.126533

Pawlicki M, Pawlicka A, Uccello F, Szelest S, D'Antonio S, Kozik R, Choraś M (2024) Evaluating the necessity of the multiple metrics for assessing explainable AI: a critical examination. Neurocomputing 602:128282. https://doi.org/10.1016/j.neucom.2024.128282

Retzlaff CO, Angerschmid A, Saranti A, Schneeberger D, Röttger R, Müller H, Holzinger A (2024) Post-hoc vs ante-hoc explanations: xAI design guidelines for data scientists. Cogn Syst Res 86:101243. https://doi.org/10.1016/j.cogsys.2024.101243

Ribeiro M, Singh S, Guestrin C (2016) Why should I trust you?: Explaining the predictions of any classifier. In: 2016 conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations, San Diego

Saeed W, Omlin C (2023) Explainable AI (XAI): a systematic meta-survey of current challenges and future opportunities. Knowl-Based Syst 263:110273. https://doi.org/10.1016/j.knosys.2023.110273

Saha D, Manickavasagan A (2021) Machine learning techniques for analysis of hyperspectral images to determine quality of food products: a review. Curr Res Food Sci 4:28–44. https://doi.org/10.1016/j.crfs.2021.01.002

Sarker IH, Janicke H, Mohsin A, Gill A, Maglaras L (2024) Explainable AI for cybersecurity automation, intelligence and trustworthiness in digital twin: methods, taxonomy, challenges and prospects. ICT Express 10(4):935–958. https://doi.org/10.1016/j.icte.2024.05.007

Sharma DK, Mishra J, Singh A, Govil R, Srivastava G, Lin JC-W (2022) Explainable artificial intelligence for cybersecurity. Comput Electr Eng 103:108356. https://doi.org/10.1016/j.compeleceng.2022.108356

Snyder H (2019) Literature review as a research methodology: an overview and guidelines. J of Bus Res 104:333–339. https://doi.org/10.1016/j.jbusres.2019.07.039

Sovrano F, Vitali F, Palmirani M (2021) Making things explainable vs explaining: requirements and challenges under the GDPR. https://doi.org/10.1007/978-3-030-89811-3_12. arXiv:2110.00758

Srinivasu PN, Sandhya N, Jhaveri RH, Raut R (2022) From blackbox to explainable AI in healthcare: existing tools and case studies. Mob Inf Syst 2022:1–20. https://doi.org/10.1155/2022/8167821

Srivastava G, Jhaveri RH, Bhattacharya S, Pandya S, Rajeswari Maddikunta PKR, Yenduri G, Hall JG, Alazab M, Gadekallu TR (2022) XAI for cybersecurity: state of the art, challenges, open issues and future directions. arXiv:2206.03585

Swamy V, Frej J, Käser T (2023) The future of human-centric eXplainable artificial intelligence (XAI) is not post-hoc explanations. arXiv:2307.00364

Tjoa E, Guan C (2021) A survey on explainable artificial intelligence (XAI): toward medical XAI. IEEE Trans Neural Netw Learn Syst 32(11):4793–4813. https://doi.org/10.1109/TNNLS.2020.3027314

Yan F, Wen S, Nepal S, Paris C, Xiang Y (2022) Explainable machine learning in cybersecurity: a survey. Int J Intell Syst 37(12):12305–12334. https://doi.org/10.1002/int.23088

Yang W, Wei Y, Wei H, Chen Y, Huang G, Li X, Li R, Yao N, Wang X, Gu X, Amin MB, Kang B (2023) Survey on explainable AI: from approaches, limitations and applications aspects. Hum-Centric Intell Syst 3(3):161–188. https://doi.org/10.1007/s44230-023-00038-y

Yensen J (2013) PICO search strategies. Online J Nurs Inform 17(3). http://ojni.org/issues/?p=2860. Accessed 25 June 2024

Yurrita M, Balayn A, Gadiraju U (2023) Generating process-centric explanations to enable contestability in algorithmic decision-making: challenges and opportunities. arXiv:2305.00739

Zhang Z, Hamadi HA, Damiani E, Yeun CY, Taher F (2022) Explainable artificial intelligence applications in cyber security: state-of-the-art in research. IEEE Access 10:93104–93139. https://doi.org/10.1109/ACCESS.2022.3204051

Zhou R, Hu T (2023) Evolutionary approaches to explainable machine learning. arXiv:2306.14786

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.